Reciprocal exchange fixes relative mechanical scales

Two measured acceleration responses determine an inertia ratio when both arise from a shared conservative spring. A connected network determines all such ratios if its cycle products are consistent. One common multiplier of inertias, spring constants, energies and actions remains free. Reciprocal exchange therefore supplies a relative-normalization mechanism, with an explicit residual absolute-scale freedom.

Status: Q06 exploratory derivation, not an accepted-ledger claim. The model and written consistency review below require independent proof review and a bounded literature comparison before promotion. No novelty claim is made.

1. Specify the response before choosing masses

Let two coordinates $x, y \in \mathbb{R}$ have the same length units and share physical time t. Suppose their measured equations are

$$\ddot{x} = -\alpha(x - y), \qquad \ddot{y} = \beta(x - y), \qquad \alpha, \beta > 0. \tag{1}$$

Both coefficients have units time^{-2}. Isolated free motion would leave the sectors' positive inertias m_1, m_2 independently arbitrary. Now require the specific mechanical completion

$$L = \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2\dot{y}^2 - \tfrac{1}{2}k(x - y)^2, \qquad k > 0. \tag{2}$$

Here masses have mass units and k has mass/time^2 units. This assumes a diagonal constant kinetic energy and one reciprocal position-dependent interaction. It is a physical model premise, not a consequence of observing arbitrary coupled motion. Its forces are $F_1 = -k(x - y)$ and $F_2 = k(x - y)$. The Euler–Lagrange equations agree with (1) exactly when

$$m_1\alpha = m_2\beta = k, \qquad \frac{m_2}{m_1} = \frac{\alpha}{\beta}. \tag{3}$$

Thus fixed responses admit precisely the positive family $(m_1, m_2, k) = m_1(1, \alpha/\beta, \alpha)$, with the entries interpreted in their stated units. Coupling constrains relative normalization; equal inertias follow only if the two acceleration responses are equal.

2. Energy exchange and the remaining action freedom

Writing $r = x - y$, the conserved energy and total momentum are

$$H = \tfrac{1}{2}m_1\dot{x}^2 + \tfrac{1}{2}m_2\dot{y}^2 + \tfrac{1}{2}kr^2, \qquad P = m_1\dot{x} + m_2\dot{y}. \tag{4}$$

Indeed $\dot{T}_1 = -kr\dot{x}$, $\dot{T}_2 = kr\dot{y}$, and $\dot{V} = kr(\dot{x} - \dot{y})$, so $\dot{H} = 0$; the two forces give $\dot{P} = 0$. To display an exchange current, assign half the interaction energy to each sector: $E_1 = T_1 + V/2$, $E_2 = T_2 + V/2$. Then

$$\dot{E}_1 = -\tfrac{k}{2}r(\dot{x} + \dot{y}) = -\dot{E}_2. \tag{5}$$

The half split is an accounting convention; conservation of H is independent of it. Generic initial data give nonzero exchange. Measuring power in an independently calibrated energy unit would add information beyond (1).

For every dimensionless $\eta > 0$, replace (m_1, m_2, k) by $(\eta m_1, \eta m_2, \eta k)$. Equations (1), all coordinate histories with the same initial positions and velocities, and their periods remain unchanged. L, H, P, E_i and the currents in (5) scale by η. So does the trajectory action $S = \int L\,dt$ on any fixed time interval. Independent rescaling of only one sector generally violates (3); common rescaling does not.

An on-shell positive action diagnostic makes this freedom visible without cancellation in S. Set $M = m_1 + m_2$, $\mu = m_1 m_2/M$ and $X = (m_1 x + m_2 y)/M$. Then

$$L = \tfrac{1}{2} M \dot{X}^2 + \tfrac{1}{2} \mu \dot{r}^2 - \tfrac{1}{2} k r^2, \qquad \omega^2 = k/\mu = \alpha + \beta. \tag{6}$$

In the stationary centre frame choose $r = A\cos(\omega t)$, $A > 0$. With $p_r = \mu \dot{r}$, its closed-orbit action is

$$I = \frac{1}{2\pi} \oint p_r \, dr = \frac{1}{2} \mu \omega A^2 = \frac{E_{\mathrm{rel}}}{\omega}. \tag{7}$$

The integral follows from $p_r \dot{r} = \mu A^2 \omega^2 \sin^2(\omega t)$ over one period. It has action units. Common rescaling sends I to ηI at fixed trajectory. Even for fixed masses and spring, the admitted small amplitudes $A \downarrow 0$ send I to zero. The full Lagrangian action over this stationary-centre period is zero because the kinetic and potential integrals coincide; (7) must not be identified with that signed integral.

The relative frequency is strictly positive for this fixed two-body model. The full system also has a free centre coordinate, so it has a zero-frequency mode. This is a finite-dimensional classical frequency statement, with no volume or continuum claim and no quantum energy gap.

3. A network consistency test

The relative mechanism extends to a finite connected undirected graph of scalar coordinates, all measured in the same length and time units. On every edge $\{i, j\}$ suppose both directed responses $a_{ij}, a_{ji} > 0$ are known:

$$\ddot{q}_i = -\sum_{j \sim i} a_{ij}(q_i - q_j). \tag{8}$$

Seek only the mechanical class

$$L = \tfrac{1}{2} \sum_i m_i \dot{q}_i^2 - \tfrac{1}{2} \sum_{\{i,j\}} k_{ij}(q_i - q_j)^2, \qquad m_i > 0, \quad k_{ij} = k_{ji} > 0. \tag{9}$$

Comparison of coefficients requires $m_i a_{ij} = m_j a_{ji} = k_{ij}$. Hence the necessary cycle condition is

$$\prod_{\ell=0}^{s-1} \frac{a_{i_\ell i_{\ell+1}}}{a_{i_{\ell+1} i_\ell}} = 1, \qquad i_s = i_0. \tag{10}$$

It is sufficient as well: choose any $m_0 > 0$, transport masses along a spanning tree using $m_j/m_i = a_{ij}/a_{ji}$, and apply (10) on each remaining edge's fundamental cycle. This makes its transported ratio agree with the edge ratio. All masses are positive and define the required springs. Any two solutions have the same mass ratio along every edge; connectivity makes their quotient one common constant. A disconnected graph instead retains one constant per connected component.

For a concrete failed test take a triangle with $a_{12} = 2\gamma$ and all other directed edge coefficients $\gamma > 0$. The cycle $1 \to 2 \to 3 \to 1$ has product 2. The required ratios would give $m_2 = 2m_1$, $m_3 = m_2$, and $m_1 = m_3$, a contradiction. Such responses cannot have completion (9). This rejects that diagonal-inertia reciprocal-spring model; it does not exclude completions with additional variables or a different kinetic form.

4. Decision and written consistency review

The positive outcome is a concrete relative calibration mechanism: reciprocal exchange joins formerly independent normalizations. Its cycle condition is a falsifiable constraint on measured responses. It does not dynamically drive masses toward a ratio; it identifies which fixed masses are compatible with the assumed energy and observed equations. A calibrated reference mass, spring force or energy transfer would fix the common multiplier empirically. It would supply the missing normalization as an input.

The coordinator's written checks are (3) by direct variation, (4)–(5) by the displayed work balance, (6)–(7) by the centre/relative change of coordinates and period integral, and (10) by path consistency. Scaling changes mechanical parameters rather than coordinate units: a fixed external force or mass standard would distinguish the models. Positive η and positive amplitudes approach zero action without using a degenerate mass endpoint. No numerical or symbolic verification scripts were used.

Q05 left open whether mutual interaction could remove its common action coefficient. Within (2) and (9), fixed mutual responses remove relative freedoms but retain the common one. Park further spring-network variants unless a concrete measured response or proposed interaction violates this completion and changes a named physical premise. For the quantum track, the next distinct test is global spin-action patching: distinguish classical symplectic consistency from consistency of a proposed phase $\exp(iS/K)$, and locate which premise, if any, forces integrality or fixes K.

Source context is the Q05 normalization calculation and conditional composition result. Their source audits are retained for their own claims. The present elementary mechanical construction has no new prior-art comparison yet; this exploratory note is not a ledger promotion.